

E185B Final Report

Project 2: Replacement of Styrofoam to a Paper-based Material for Shipment

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Introduction

Background

EPS (Polystyrene foam, Expanded Polystyrene) is commonly used for packaging across industries, and it has increasingly been shown to be environmentally harmful. While EPS is cheap and very effective at protecting products, it is not compostable or recyclable. When disposed of, it quickly disintegrates into microplastics that can leach into our waterways and food supplies, with known carcinogens, posing a threat to public health. As consumers increasingly prioritize sustainability, companies are adapting to changing consumer desires. A study of consumers across the US, Netherlands, and UK found that 70% of consumers make purchasing decisions based on the sustainability of the packaging.¹

One promising alternative to EPS is molded pulp, a material made from recycled paper fibers that is fully compostable, recyclable, and free from harmful chemicals. Molded pulp can be formed into complex, custom shapes and offers competitive protection and cost compared to EPS. Initially, our project explored the use of molded pulp as a direct replacement for Styrofoam in shipping applications.

However, based on client feedback emphasizing the need for simpler, more cost-effective manufacturing processes, we pivoted to using folded cardboard structures instead of molded pulp. Cardboard offers high sustainability, wide recyclability, and effective protective properties when engineered correctly. Our solution uses several types of cardboard-based protection:

- **Cardboard Corners:** We designed foldable, kirigami-inspired cardboard corner structures that can flexibly fit into a variety of box sizes while providing cushioning and stability at the corners of the packaged product.

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<https://www.flexcon.com/press-releases/new-data-reveals-70-of-consumers-choose-products-based-on-sustainability#:~:text=New%20Data%20Reveals%2070%25%20of%20Consumers%20Choose%20Products%20Based%20on%20Sustainability,-Date%3A%20August%202013>

- **Firm Cross-Sections:** For additional support, we stacked and glued layers of corrugated cardboard to create dense, rigid barriers along the sides of the record player. These firm cross-sections reinforce the structure against lateral impacts.
- **Cardboard Mesh (Woven Cardboard):** At the top and bottom of the record player, we implemented a woven cardboard mesh that distributes weight and impact forces while providing shock absorption.
- **Cardboard Barriers:** To prevent shifting within the box, we inserted additional cardboard barriers between the corners and the product, effectively filling empty spaces and adding another layer of protection.

This approach allowed us to meet the project's sustainability, protection, and cultural alignment goals while streamlining the manufacturing process for ease of scaling. In addition to sustainability, we focused on designing for dual-purpose use: the foldable cardboard corners were designed with slits to allow users to repurpose them as record display stands, extending the life of the packaging materials through upcycling. The inclusion of a Japanese *tenugui* towel as a protective cover to replace the plastic cover further emphasized cultural heritage by honoring the origins of the Audio-Technica brand, creating a packaging experience that felt both thoughtful and authentic. Photographs of each component, including the cardboard mesh, firm cross-sections, foldable corners, and gap-filling barriers, are included in the following sections for reference.



Figure 1. Cardboard Corner Protection

Foldable kirigami-inspired cardboard corners designed to cushion and stabilize the record player at all four box corners.



Figure 2. Firm Cross-Section Barriers

Stacked and glued corrugated cardboard sections placed along the sides of the record player to provide reinforced lateral impact protection.

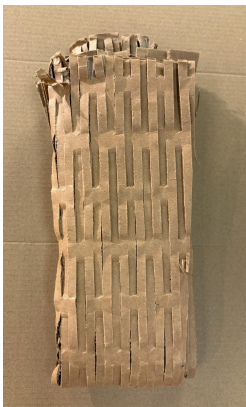


Figure 3. Woven Cardboard Mesh

Interwoven strips of cardboard placed above and below the record player; distributing impact forces and providing shock absorption during shipping.



Figure 4. Cardboard Gap-Filling Barriers

Inserted cardboard pieces that fill the voids between the product and box corners, preventing movement and adding additional structural support.



Figure 5. Tenugui Towel Protective Covering

Traditional Japanese tenugui cloth used to replace the plastic foam sheet, providing a protective layer against scratches while enhancing cultural branding.



Figure 6. Full Packaging Assembly

Complete view of the final packaging system inside the box, showing the integration of cardboard corners, cross-sections, woven mesh, and barriers.

Problem Statement

Audio-Technica currently uses EPS (Polystyrene foam, Expanded Polystyrene) packaging to ship their products. EPS is very moldable and inexpensive, and protects products from damage when being shipped very effectively. However, the production of EPS is very harmful to the environment, and it is not reusable. As a result, Audio-Technica is looking for a reusable, paper-based, and affordable alternative to EPS packaging.

Deliverables

- The team must design a reusable and affordable packing alternative to EPS packaging that Audio-Technica would have the ability to implement.

Functions

- Customizable to any Audio-Technica product
- Protects Audio-Technica products from being damaged under standard shipping conditions

Constraints

- Must be paper-based, including cardboard
- Must pass Audio-Technica's material test standards
 - 20 lbs product of 6 faces (tetrahedron) at 70cm dropping 6 times at each of the six vertices and remaining intact
- Must be recyclable and/or biodegradable

Objectives

- Be competitive in cost with EPS
 - The cost of the packaging product should not be excessively more expensive than EPS packaging
- Be mass producible
 - Audio-Technica should be able to quickly produce mass amounts of the packaging product

Design Process

Design Approach

Our design approach was centered around three goals: sustainability, dual-purpose packaging, and cultural heritage. We knew we wanted to stick to cardboard to make our packaging sustainable. In addition to making a protective and sustainable packaging design, we wanted to ensure that some parts of the packaging could be upcycled once the turntable was removed from the box, so we made the protective corners have dual functionality as record holders. Finally, we wanted to honor the Japanese cultural heritage of Audio-Technica and incorporated a Japanese cotton textile to protect the glass part of the record player from any scratches.

Our design process began with exploring various fabrication techniques, including corrugation manipulation, kirigami, molded cardboard pulp, and woven strategies. While each method presented unique advantages, we decided to pursue corrugation manipulation and folding techniques as these two methods use standard cardboard, which is the most cost-effective and easiest to implement. In addition to meeting the primary packaging requirements, we sought to push beyond the problem statement by incorporating a secondary function into our design. This led us to develop a concept where the folded cardboard corners could double as record stands, adding an innovative and multi-use aspect to the packaging solution.

We sought to use cardboard folding to make folded corners that would fit snugly to the record player, holding it in place and protecting the most vulnerable edges. Then, manipulating the cardboard corrugation and orienting it to provide strength in the critical directions, we created a border in the Audio-Technica box that would hold the corners and record player in place. Finally, to offer extra protection, we added a woven cardboard material folded into a cushion above and below the record player.

Research

Through drop testing, we were able to identify key weaknesses in our first prototype and think that they could easily be fixed with a little more targeted reinforcement. We believe that the cardboard mesh material used to cushion the top and bottom of the record player could be used to wrap the entire apparatus, helping to cushion the sides better. We identified the hinges as the

most fragile parts of the record player and would implement more reinforcement on that side of the packaging if given the opportunity to make another prototype.

Prototypes

Our first prototype was made out of molded cardboard pulp. After consulting multiple guides, we 3D printed a geometric mold for the cardboard mixture to demonstrate the precise angles and strength that even a homemade mold can achieve. To make the molded pulp, we blended cardboard/paper, a type of starch, and water. Once the mixture dried, we found that experimenting with different ratios and paper types yielded different strengths. We discovered that tapioca starch was not a viable binder and opted for cornstarch instead. We also learned that a finer pulp mixture was stronger and easier to shape, which could be achieved through a longer blending time. Ultimately, we found that slowly adding the cardboard to a pre-blended mixture of warm water and cornstarch yielded the strongest form.

When we realized this would not be easy to manufacture for Audio-Technica, we pivoted, focusing on techniques that would manipulate standard cardboard. Our second Prototype Figure 6 showcases the folded cardboard corners and manipulated corrugated cardboard. By cutting thin strips of single and triple-walled cardboard and layering them together, we were able to create a rigid and strong surrounding for the record player.

We also wanted to create snug corners for the record player and thought this could be achieved with precise paper folding techniques similar to kirigami. We started with paper to perfect the template. Once we had the dimensions figured out, we cut the same shape out of cardboard and folded it into the corners for the record player. Finally we included slits so that they could be used to display records by the turntable owner.

Finally, we wrapped the plexiglass part of the turntable with a traditional Japanese cotton textile, taking inspiration from the tradition of furoshiki. The final Prototype can be seen in Figure 6 of the introduction.

Testing

Motivation

Figure 7 shows the AudioTechnica turntable that served as the basis for our packaging design

and was later used for drop testing. All packaging designs created by the team were custom-fit to this turntable model; however, the packaging designs should be applicable to other turntable models of different sizes. The turntable used weighs 5.73 lbs, so we drop tested from 8.0 ft (2.44 m) packaging to meet Audio-Technica's testing standards, dictated by the law of Energy Conservation Appendix A.



Figure 7. Audio-Technica At-LP60X-BW model used for drop testing

Testing Procedure

Audio-Technica's drop testing standards dictate that drop tests should be conducted using at least a 20-lb object from a height of 70 cm. Our turntable was 5.73 lbs, so using the principles of gravitational potential energy Appendix A, we found that if the packaged turntable is dropped from a height of 8.0 ft (2.44 m), it should withstand the equivalent potential energy as a 20 lbs object dropped from a height of 70 cm. Once the turntable was properly packaged, we conducted one round of drop tests, which consisted of drop testing the packaged turntable on each of its faces, from 70 cm (approximately 28 inches), to check if there was any damage. We then conducted two rounds of drop testing from a height of 8 ft (2.44 m).

Results

We found the drop tests from 70 cm to be mostly successful, however, a plastic hinge broke off as displayed in Appendix D when dropped such that the opposite side of the packaging landed first. We believe this is due to the fact that there was a gap between the plastic casing and supportive cardboard barrier, causing the turntable not to be held in place, thus moving around more within the packaging upon impact. We made this design decision because we hypothesized that if the plastic casing was in contact with the supportive cardboard barrier, such that the cardboard barrier's firm cross-section was oriented towards the turntable, the barrier may have been too firm a support for the plastic, making it more vulnerable to cracking upon impact. After conducting that first round of testing, we conducted two full rounds of drop testing from 8 ft (2.44 m). We stopped testing after two rounds because the packaging and turntable had received a notable amount of damage displayed in Appendix E; however, the damage dealt to the turntable was largely along the clear plastic casing and plastic hinges. The turntable took minor damage to no damage at all when dropped, such that the side with the plastic casing impacted the ground first, as well as when the side opposite to the plastic casing impacted the ground first. Those two sides were the sides protected by the woven cardboard mesh, indicating that the woven cardboard mesh is a promising material to use for sustainable packaging. However, the other sides were protected by firm cross-sectional barriers and cardboard corners, which proved not to be as protective as the woven cardboard mesh. We found that those sides resulted in substantial cracking, and the plastic casing and some minor components of the turntable broke after two rounds of drop testing on those sides. The firm cross-sectional barriers and cardboard corners provided excellent stability and multi-use functionality, so perhaps reinforcing them with woven cardboard mesh would have yielded much more successful results.

Conclusion/Future Work

Although our final design did not pass Audio-Technica's full drop testing standards, our design process revealed valuable insights for creating the future of sustainable packaging. Throughout our project, we grappled with the concept of "sustainability" in a world of mass production. To maximize sustainability and significantly cut down on packaging, we identified that we needed to design reusable packaging that people would want to keep, and more importantly, *would not want to throw away*. Our cardboard engineered solution used kirigami-inspired folded corners, layered corrugated barriers, woven mesh, and internal gap fillers. By testing a variety of cardboard manipulation techniques, we hope to provide Audio-Technica with insight for future sustainable packaging designs.

While we found that our packaging was initially successful for the first drop tests, continued testing revealed key shock absorption weaknesses in the design. With further iteration and specific damage analysis, these weaknesses could be eliminated for optimized product protection. Despite the performance limitations, the core concept of a paper-based, recyclable packaging system demonstrated potential as a sustainable alternative to EPS. Our design incorporated aesthetic elements and crafted a brand-aligned narrative reflective of Audio-Technica's culture and core values. Our versatile prototype balances customization, production efficiency, and cost. With further strengthening to create a more durable iteration, this sustainable design alternative to Styrofoam would allow Audio-Technica to maintain the quality and protection of their audio gear while demonstrating their commitment to reducing environmental impact.

Appendix

Appendix A - Height Calculation Using Gravitational Potential Energy:

New height Calculation Using Gravitational Potential Energy:

$$m_1 g h_1 = m_2 g h_{\text{new}}$$

$$\left(\frac{20}{2.2} \text{ kg}\right) (9.81 \text{ m/s}^2) (0.7 \text{ m}) = \left(\frac{5.73}{2.2} \text{ kg}\right) (9.81 \text{ m/s}^2) h_{\text{new}}$$

$$h_{\text{new}} = \frac{\left(\frac{20}{2.2} \text{ kg}\right) (9.81 \text{ m/s}^2) (0.7 \text{ m})}{\left(\frac{5.73}{2.2} \text{ kg}\right) (9.81 \text{ m/s}^2)}$$

$$h_{\text{new}} \approx 7.44 \text{ m} \approx 8.0 \text{ ft}$$

Appendix B - 70 cm Drop Tests:



Appendix C - 8 ft (2.44 m) Drop Tests:



Appendix D - Damage After 70 cm Drop Tests:



Appendix E - Total Damage After All Drop Tests:

